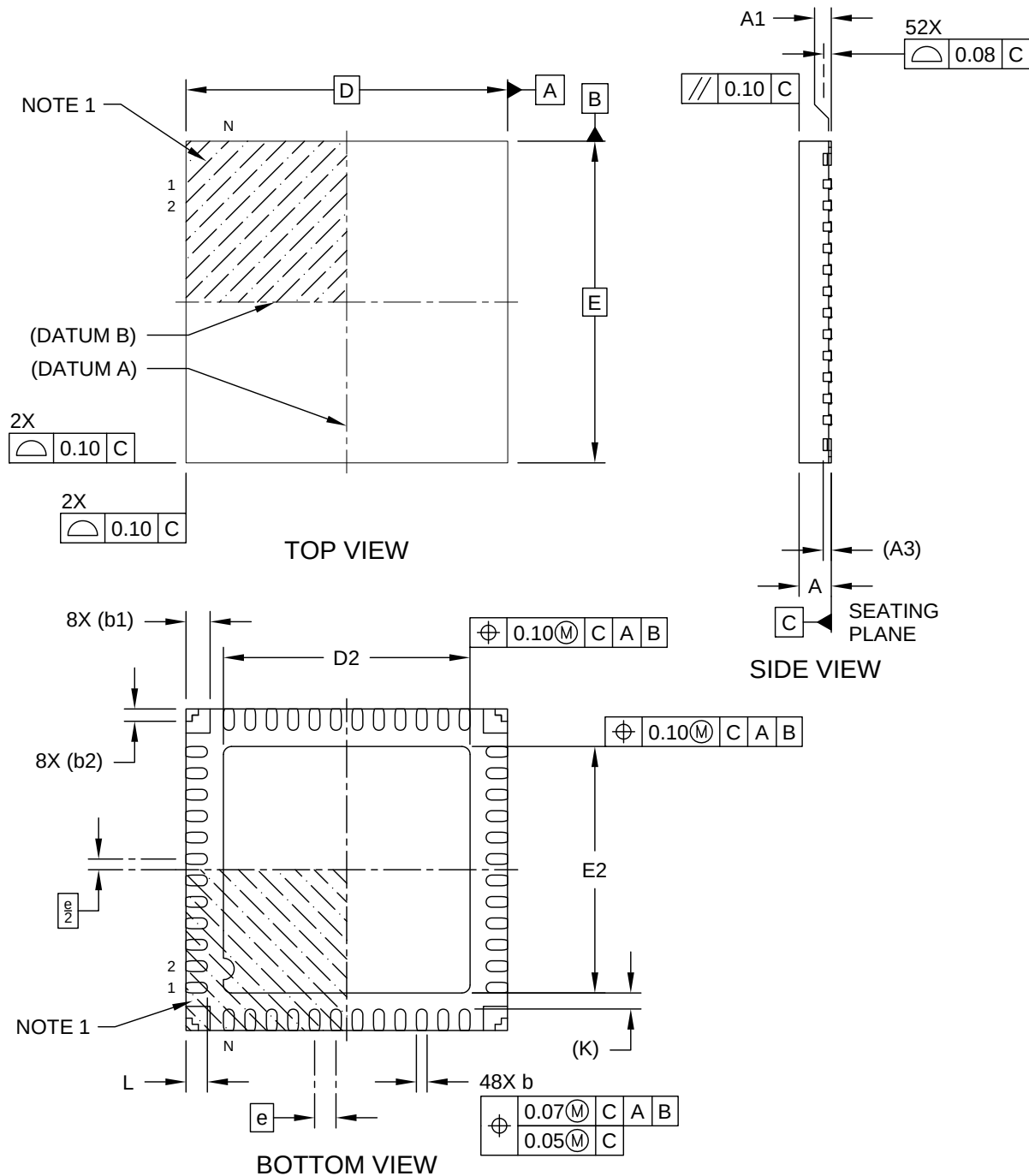


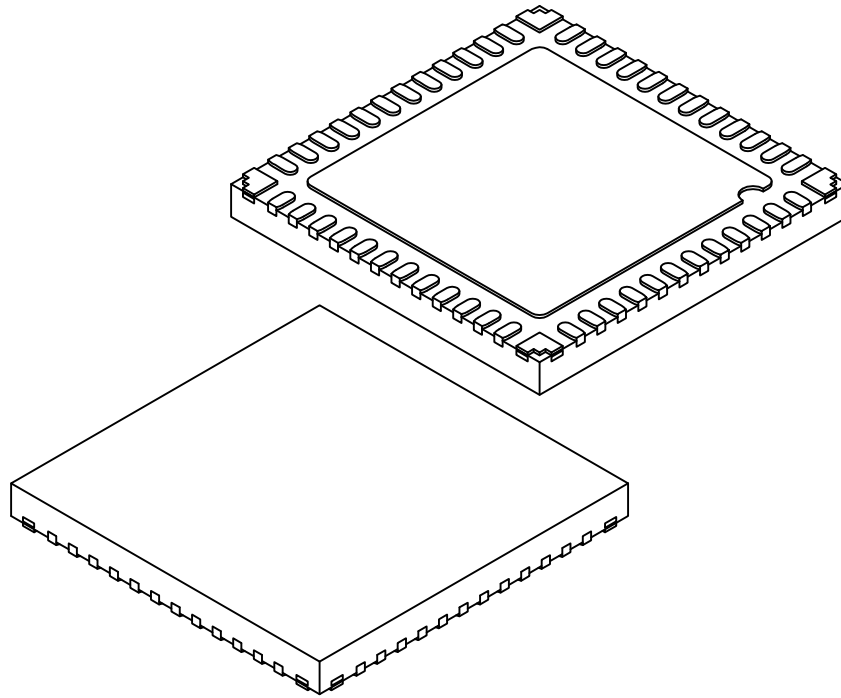
48-Lead Ultra Thin Plastic Quad Flat, No Lead Package (PTX) - 6x6 mm Body [UQFN] With Corner Anchors and 4.6x4.6 mm Exposed Pad

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



48-Lead Ultra Thin Plastic Quad Flat, No Lead Package (PTX) - 6x6 mm Body [UQFN] With Corner Anchors and 4.6x4.6 mm Exposed Pad

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



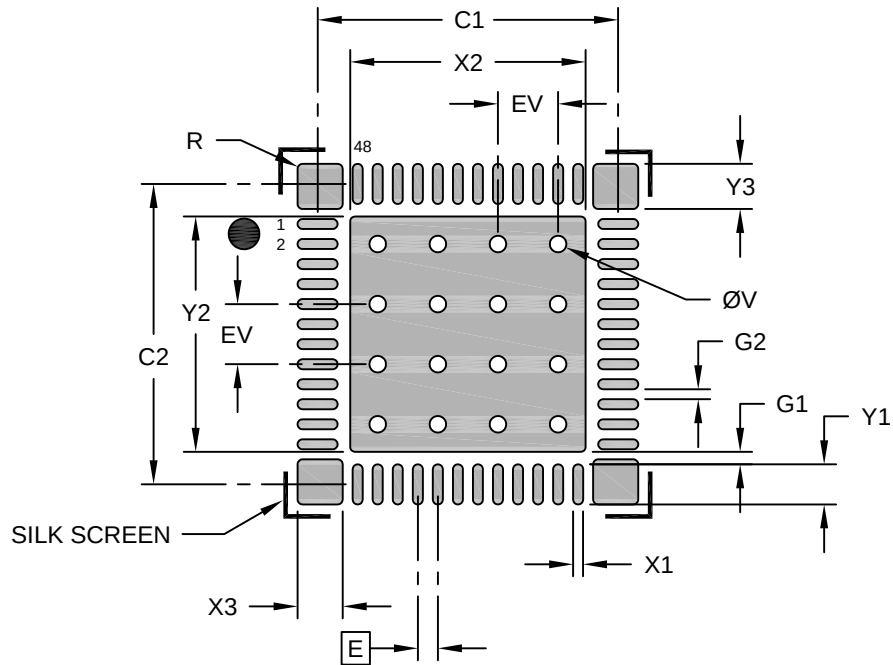
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	48		
Pitch	e	0.40 BSC		
Overall Height	A	0.50	0.55	0.60
Standoff	A1	0.00	0.02	0.05
Terminal Thickness	A3	0.15 REF		
Overall Length	D	6.00 BSC		
Exposed Pad Length	D2	4.50	4.60	4.70
Overall Width	E	6.00 BSC		
Exposed Pad Width	E2	4.50	4.60	4.70
Terminal Width	b	0.15	0.20	0.25
Corner Anchor Pad	b1	0.45 REF		
Corner Anchor Pad, Metal-free Zone	b2	0.23 REF		
Terminal Length	L	0.35	0.40	0.45
Terminal-to-Exposed-Pad	K	0.30 REF		

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

48-Lead Ultra Thin Plastic Quad Flat, No Lead Package (PTX) - 6x6 mm Body [UQFN] With Corner Anchors and 4.6x4.6 mm Exposed Pad

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

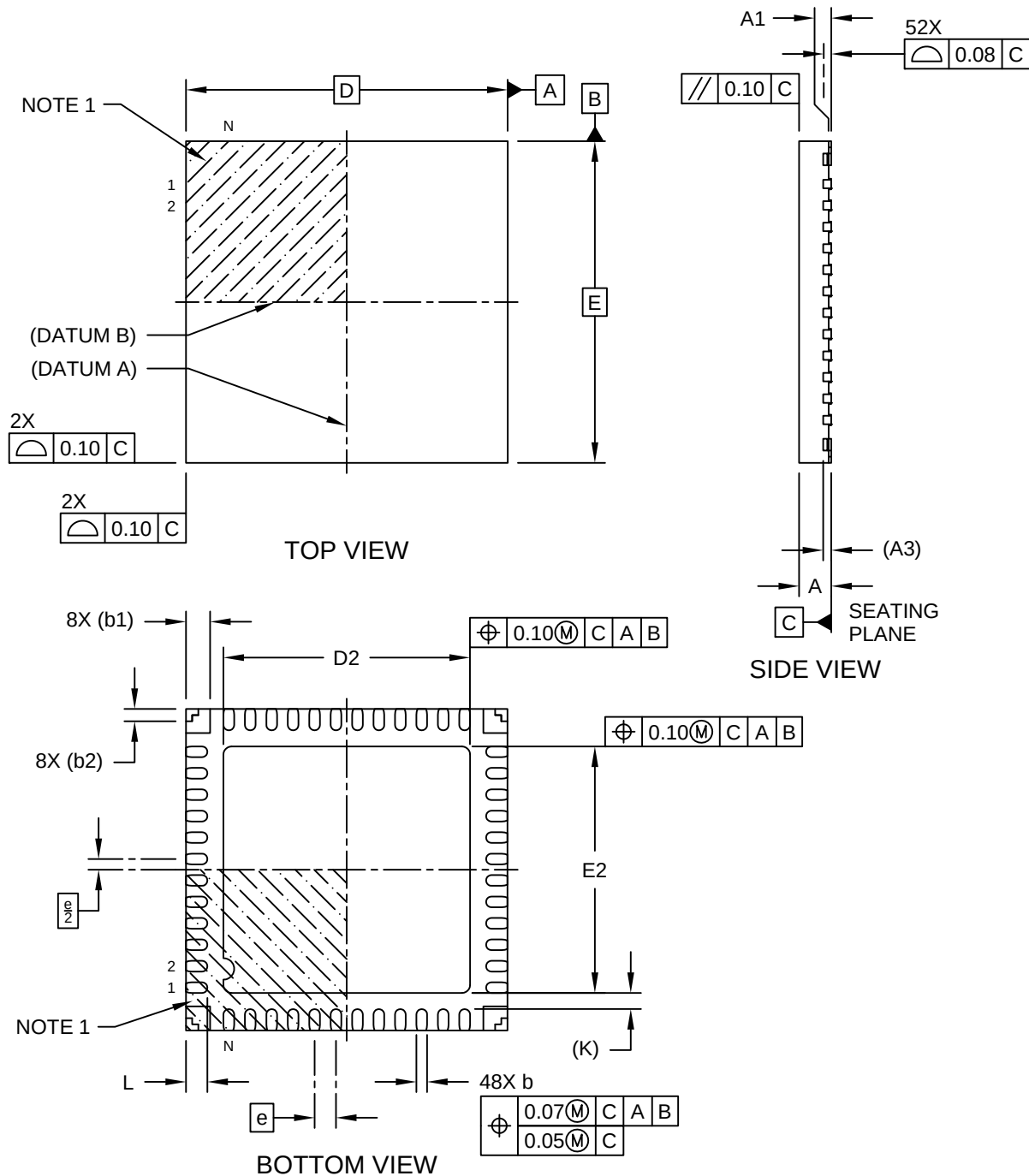
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.40 BSC		
Center Pad Width	X2			4.70
Center Pad Length	Y2			4.70
Contact Pad Spacing	C1		6.00	
Contact Pad Spacing	C2		6.00	
Contact Pad Width (X48)	X1			0.20
Contact Pad Length (X48)	Y1			0.80
Corner Anchor Pad Width (X4)	X3			0.90
Corner Anchor Pad Length (X4)	Y3			0.90
Pad Corner Radius (X 20)	R			0.10
Contact Pad to Center Pad (X48)	G1	0.25		
Contact Pad to Contact Pad	G2	0.20		
Thermal Via Diameter	V		0.33	
Thermal Via Pitch	EV		1.20	

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

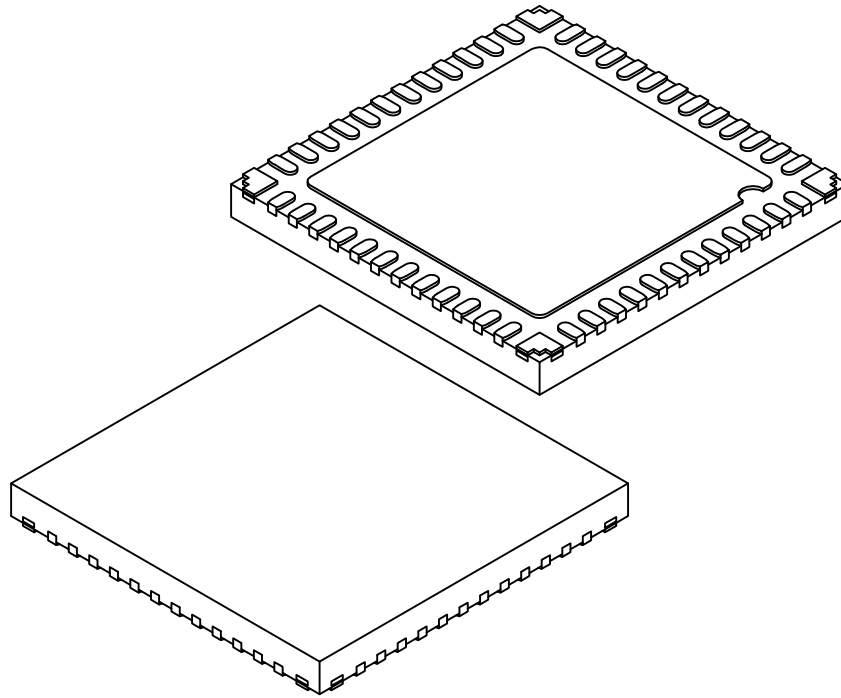
48-Lead Ultra Thin Plastic Quad Flat, No Lead Package (M4) - 6x6 mm Body [UQFN] With Corner Anchors and 4.6x4.6 mm Exposed Pad

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



48-Lead Ultra Thin Plastic Quad Flat, No Lead Package (M4) - 6x6 mm Body [UQFN] With Corner Anchors and 4.6x4.6 mm Exposed Pad

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



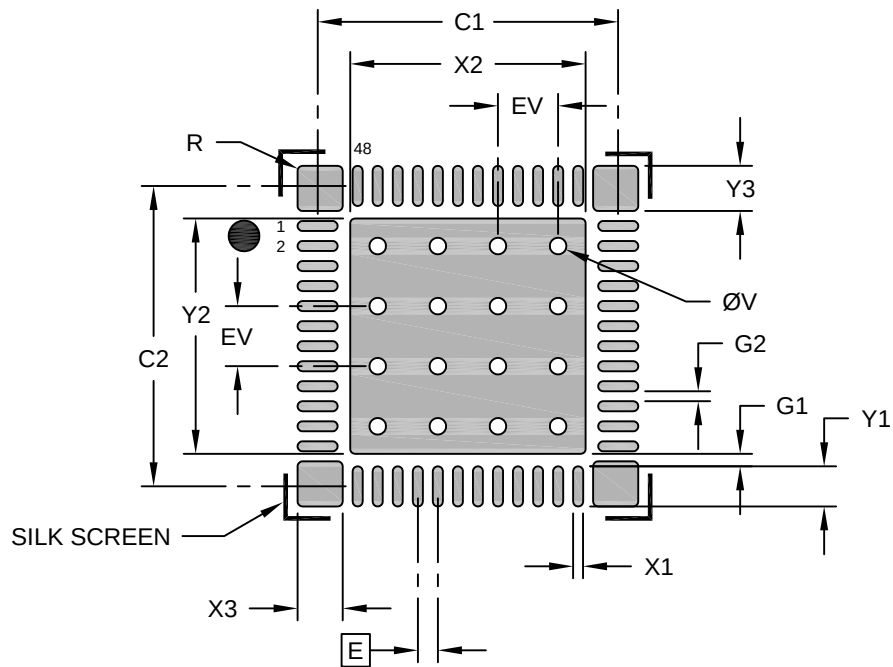
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	48		
Pitch	e	0.40 BSC		
Overall Height	A	0.50	0.55	0.60
Standoff	A1	0.00	0.02	0.05
Terminal Thickness	A3	0.15 REF		
Overall Length	D	6.00 BSC		
Exposed Pad Length	D2	4.50	4.60	4.70
Overall Width	E	6.00 BSC		
Exposed Pad Width	E2	4.50	4.60	4.70
Terminal Width	b	0.15	0.20	0.25
Corner Anchor Pad	b1	0.45 REF		
Corner Anchor Pad, Metal-free Zone	b2	0.23 REF		
Terminal Length	L	0.35	0.40	0.45
Terminal-to-Exposed-Pad	K	0.30 REF		

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

**48-Lead Ultra Thin Plastic Quad Flat, No Lead Package (M4) - 6x6 mm Body [UQFN]
With Corner Anchors and 4.6x4.6 mm Exposed Pad**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.40 BSC		
Center Pad Width	X2			4.70
Center Pad Length	Y2			4.70
Contact Pad Spacing	C1		6.00	
Contact Pad Spacing	C2		6.00	
Contact Pad Width (X48)	X1			0.20
Contact Pad Length (X48)	Y1			0.80
Corner Anchor Pad Width (X4)	X3			0.90
Corner Anchor Pad Length (X4)	Y3			0.90
Pad Corner Radius (X 20)	R			0.10
Contact Pad to Center Pad (X48)	G1	0.25		
Contact Pad to Contact Pad	G2	0.20		
Thermal Via Diameter	V		0.33	
Thermal Via Pitch	EV		1.20	

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
2. For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process